A Project Report on

COMMUNITY EVENT MANAGEMENT SYSTEM

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "**Community Event Management System"** is a bonafide work of **Harshita Waghare (24106038), Jagruti Singh (24106147), Riya Singh (24106027)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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ABSTRACT

The Community Event Management System is a web-based application designed to simplify and streamline the process of organizing community events. Traditional event management methods often rely on manual processes, which lead to inefficiencies such as poor coordination, lack of communication, and resource mismanagement.

This system provides a centralized platform where administrators can create and manage events, volunteers can register and participate in assigned tasks, and participants can easily browse and register for events. The platform integrates features such as task assignment, real time tracking, communication, and budget management.

The system is developed using Flask for backend processing, React for frontend interaction, and MySQL/PostgreSQL for database management. The implementation demonstrates improved efficiency, better volunteer coordination, and enhanced user experience.

This project contributes to Sustainable Development Goal 11 by promoting organized and inclusive community participation through digital transformation.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

ML – Machine Learning

SDG – Sustainable Development Goal

CEMS – Community Event Management System

DB – Database

UI – User Interface

API – Application Programming Interface

HTML – Hypertext Markup Language

CSS – Cascading Style Sheets

SQL – Structured Query Language

Chapter 1 Introduction

1. Introduction

1.1 Background

Community events are essential for social interaction, awareness programs, and local development. However, organizing such events requires proper planning, coordination, and resource management. Many organizations still rely on manual methods such as spreadsheets and messaging apps, which are inefficient and error-prone.

1.2 Existing Challenges

The following challenges are commonly observed in traditional event management approaches:

- Lack of centralized management system
- Poor communication between organizers and volunteers
- Difficulty in tracking tasks and responsibilities
- Inefficient participant registration process

1.3 Motivation

The motivation behind this project is to create a digital solution that enhances community participation and simplifies event management. Inspired by SDG Goal 11, the system aims to build sustainable and organized communities through the use of modern web technologies.

1.4 Objectives

The primary objectives of this project are as follows:

- Develop a centralized event management system
- Provide role-based access for Admin, Volunteer, and Participant
- Enable efficient task assignment and tracking
- Improve communication and coordination among users

Chapter 2 Literature Survey

2. Literature Survey

Various studies and existing platforms have contributed to the development of event management systems. This chapter reviews the existing tools and relevant research works.

2.1 Review of Existing Platforms

Eventbrite and Meetup are widely used platforms that allow users to create and manage events. However, they mainly focus on event hosting and lack features such as volunteer coordination and detailed task tracking.

Google Forms is often used for event registration but does not provide event management or analytics features.

2.2 Related Research

Research by Wang et al. (2006) highlights the importance of database connectivity using MySQL, which is essential for handling event data. Rathore et al. (2023) discuss security frameworks for web applications using Flask, emphasizing secure authentication systems. Chen et al. (2022) focus on Python-based data analysis, which can be used for event data processing.

2.3 Comparison of Existing Systems

System	Features	Advantages	Limitations
Eventbrite	Event hosting	Scalable	No volunteer module
Meetup	Community events	Social engagement	Limited customization
Google Forms	Registration	Easy to use	No tracking or analytics

Table 2.1: Comparison of Existing Event Management Systems

Chapter 3 Limitations of Existing Systems

3. Limitations of Existing Systems

The existing event management systems such as Google Forms, Eventbrite, and Meetup provide basic functionalities for event creation and participant registration. However, these systems exhibit several limitations when applied to community-level event management.

3.1 Identified Limitations

Lack of Integrated Volunteer Management: Most platforms focus only on event creation and participant registration but do not provide mechanisms for assigning tasks to volunteers or tracking their performance.

Absence of Centralized Communication: Organizers often rely on external applications such as messaging platforms or emails. This results in fragmented communication and increases the chances of missing important updates.

Limited Customization Capabilities: Platforms like Eventbrite are designed for large-scale commercial events and do not cater to the specific needs of local communities or nonprofit organizations.

Lack of Real-Time Monitoring and Analytics: Organizers are unable to track volunteer participation, event progress, or registration trends effectively.

Absence of Role-Based Access Control: Many systems do not differentiate between administrators, volunteers, and participants, which reduces system security and restricts functionality.

Data Redundancy and Inconsistency: Without proper database integration, duplicate entries and incorrect data can occur, leading to confusion and inefficiency.

No Feedback System: Most platforms do not provide a structured way to collect feedback from participants and volunteers after the event.

Scalability Issues: Many systems are not designed to handle a growing number of users or events efficiently.

3.2 Limitations Analysis Table

Limitations Analysis Table

Limitation Category	Eventbrite	Meetup	Google Forms	Impact
Volunteer Management	No	Partial	No	Poor coordination
Real-time Communication	No	No	No	Delays and errors
Analytics/Reporting	Partial	No	No	No data-driven improvements
Customization for Local Needs	Limited	Limited	No	Mismatch with requirements
Role-Based Access	No	No	No	Security risk

Table 3.1: Limitations Analysis of Existing Systems

Chapter 4 Problem Statement and Objectives

4. Problem Statement and Objectives

4.1 Problem Statement

Community events are important for social interaction and development, but their management is often inefficient due to the use of manual or partially digital systems. Existing platforms mainly focus on event registration and do not provide a complete solution for managing volunteers, tasks, and communication.

Organizers face challenges such as poor coordination, lack of real-time tracking, and difficulty in assigning and monitoring tasks. There is no centralized system that integrates all activities, leading to data inconsistency and communication gaps.

Additionally, the absence of validation mechanisms results in duplicate registrations and improper resource utilization. Therefore, there is a need for a centralized and efficient system that can manage events, volunteers, and participants in an organized manner.

4.2 Objectives

The main objective of this project is to develop a Community Event Management System that simplifies and improves event organization. The specific objectives are:

1. To develop a centralized platform for managing events
2. To implement role-based access for Admin, Volunteer, and Participant
3. To enable task assignment and tracking for volunteers
4. To simplify the participant registration process
5. To ensure data validation and prevent duplicate entries
6. To provide real-time updates and monitoring
7. To improve coordination and communication among users
8. To design a user-friendly and efficient interface

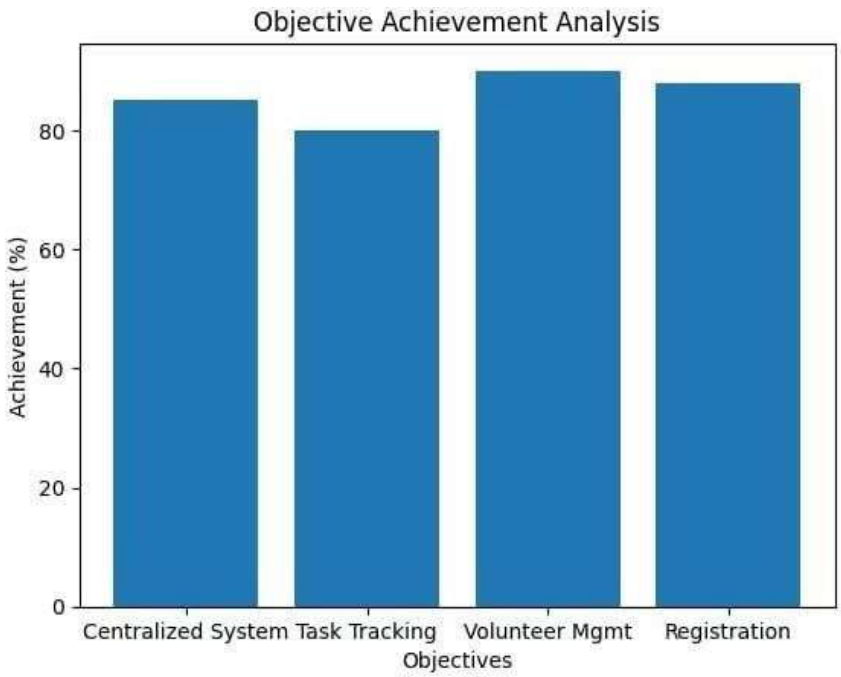

Figure 4.1: Objectives Diagram

Chapter 5 Proposed System

5. Proposed System

5.1 Overview of Proposed System

The proposed Community Event Management System is a web-based platform designed to efficiently manage community events by integrating event creation, volunteer coordination, and participant registration into a centralized system. The system aims to overcome the limitations of existing platforms by providing role-based access, real-time tracking, and improved communication.

5.2 Analysis / Framework / Algorithm

The system follows a client-server architecture, where the frontend interacts with the backend through HTTP requests. The backend processes user requests and performs database operations.

5.2.1 Framework Used

- Backend: Flask (modular structure using Blueprints)
- Frontend: HTML, CSS, Bootstrap
- Database: MySQL

5.2.2 Algorithm (Working Steps)

Step 1: User accesses the system through the web interface
Step 2: System verifies user role (Admin / Volunteer / Participant)
Step 3: Admin creates event and assigns tasks
Step 4: Volunteers select or are assigned tasks
Step 5: Participants register for events
Step 6: System validates data and stores it in database
Step 7: Real-time updates are displayed on dashboards

5.3 Design Details

5.3.1 System Modules

Admin Module:

- Create, update, and delete events
- Assign tasks to volunteers
- Monitor registrations

Volunteer Module:

- Register / Login

- View assigned tasks
- Update task status

Participant Module:

- View events
- Register for events
- Access event details

5.3.2 Database Design

- Tables: Users, Events, Tasks, Registrations, Volunteers
- Use of Primary Keys and Foreign Keys for relationships
- Ensures data integrity and avoids redundancy

5.4 Methodology

1. Requirement Analysis: Identified problems in existing systems such as lack of coordination and manual processes.
2. System Design: Designed system architecture and database schema to support multiple user roles.
3. Implementation: Developed backend using Flask and connected it with MySQL database. Frontend was built using Bootstrap for better user experience.
4. Testing: Tested system functionality, data validation, and user interactions to ensure reliability.
5. Deployment: System can be deployed on a local server or cloud platform.

5.5 Advantages of Proposed System

- Centralized management of events
- Efficient volunteer coordination
- Real-time updates and monitoring
- Reduced manual effort
- Scalable and flexible design

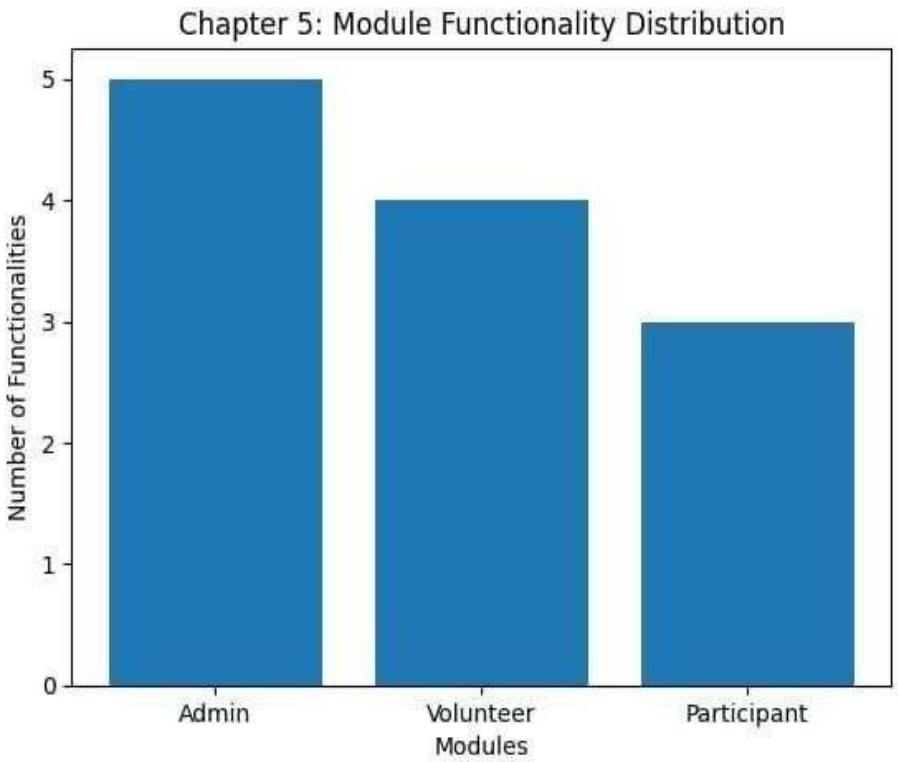

Figure 5.1: Proposed System Architecture

Chapter 6 Experimental Setup

6. Experimental Setup

6.1 Database Details

The system uses a relational database (MySQL) to store and manage all event-related data. The database includes tables for Users, Events, Tasks, Registrations, and Volunteers. All participant registration details are stored with proper validation to prevent duplicate entries. All inputs are validated before storing in the database to prevent errors and duplication.

6.2 Performance Evaluation Parameters

The performance of the system is evaluated using the following parameters:

- Response Time: Measures the time taken by the system to process user requests and display results.
- Accuracy: Ensures correct task assignment and proper handling of event registrations.
- Data Integrity: Maintains consistency and correctness of stored data using validation and constraints.
- User Satisfaction: Evaluated based on ease of use, interface design, and system efficiency.
- System Reliability: Ensures the system performs consistently without failures during multiple operations.

6.3 Software and Hardware Setup

6.3.1 Software Requirements

- Operating System: Windows / Linux / macOS
- Backend: Flask (Python Framework)
- Frontend: HTML, CSS, Bootstrap
- Database: MySQL
- Development Tools: VS Code / PyCharm
- Browser: Google Chrome / Firefox

6.3.2 Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB

- Storage: 10 GB free space
- Internet connection for testing and deployment

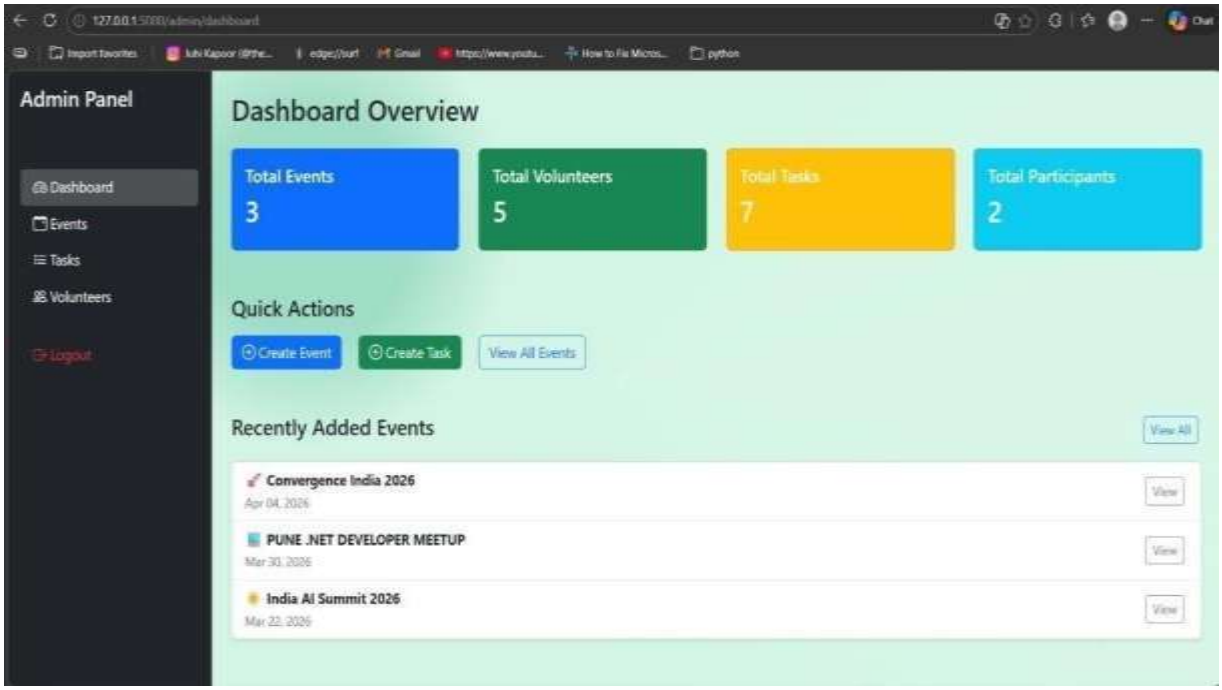

Figure 6.1: Database Schema

Chapter 7 Results and Discussion

7. Results and Discussion

7.1 Results

The Community Event Management System was successfully developed and tested under different scenarios to evaluate its performance and functionality. The system efficiently manages event creation, volunteer coordination, and participant registration. The Admin module allows users to create events, assign tasks, and monitor progress, while the Volunteer module helps in tracking assigned tasks and updating their status. The Participant module provides an easy interface for browsing and registering for events. The system also ensures data accuracy through proper validation, preventing duplicate entries. Real-time updates on dashboards further help users track event activities effectively.

7.2 Discussion

The results indicate that the proposed system effectively addresses the limitations of existing event management platforms. The inclusion of a volunteer management module is a key improvement over traditional systems, which typically focus only on event registration.The implementation of role-based access control enhances system security and ensures that users can only access relevant features. The use of database constraints and validation techniques helped in maintaining data integrity.

7.3 Comparative Analysis

Feature	Existing Systems	Proposed System
Event Management	Available	Available
Volunteer Management	Not Available	Available
Task Tracking	Limited	Fully Implemented
Real-time Updates	Limited	Available
Customization	Limited	Flexible

Table 7.1: Comparative Analysis of Proposed vs Existing Systems

7.4 Graphical Representation

Figure 7.1: System Response Time Comparison

Figure 7.2: Participant Registration Statistics

Chapter 8 Conclusion and Future Work

8. Conclusion and Future Work

8.1 Conclusion

The Community Event Management System was successfully designed and developed to address the challenges faced in organizing and managing community events. The system provides a centralized platform that integrates event creation, volunteer coordination, and participant registration into a single interface. The project effectively meets its primary objectives by implementing role-based access for administrators, volunteers, and participants. The system also ensures data integrity and accuracy through validation mechanisms and database constraints, reducing errors such as duplicate registrations and incorrect entries. In addition, the system enhances communication and coordination among users by providing real time updates and centralized access to information. The user-friendly interface ensures ease of use, making the system suitable for both technical and non-technical users.

Overall, the proposed system demonstrates a reliable, efficient, and scalable solution for community event management. It successfully overcomes the limitations of existing systems and contributes to improved organization and participation in community activities.

8.2 Future Work

Although the system performs effectively, there are several enhancements that can be implemented to improve its functionality and scalability in the future.

Integration with email and notification services to provide automatic updates to users regarding event schedules and task assignments.

Integration of a payment gateway to support paid events and online transactions.

Implementation of data analytics and reporting features to allow administrators to analyse event performance.

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